

POST-RDP EXECUTION — INCIDENT REPORT

Suspicious PowerShell Process Spawn Following Successful RDP Authentication

Project	HOME SOC LAB
Scenario	Controlled Process Execution after RDP
Status	Detection successfully validated
Test date	23 August 2026

1. Executive Summary

Following successful RDP authentication to the Windows 11 SOC host, PowerShell activity was observed in the authenticated user session. Sysmon recorded Event ID 1 (Process Create) for PowerShell, where one PowerShell process spawned another PowerShell process under the same user context. Wazuh generated Rule 92027 at level 4. The evidence supports suspicious post-RDP execution requiring investigation. The supplied telemetry does not by itself establish malicious intent

2. Lab Environment

Component	Role	Host-Only IP
Kali Linux	Attack simulation / RDP test source	192.168.56.105
Windows 11	Target / Windows Security telemetry source	192.168.56.106
Wazuh	SIEM / detection and alerting	Wazuh server in lab

Network safety: the lab uses a Host-Only adapter for the attack path between the VMs. The test was bounded to the Windows VM and was not directed at the user's host machine.

3. Attack / Lab Scenario

Successful RDP session established.

Authenticated user context: WIN11\vboxuser.

PowerShell launched within the RDP session.

A second PowerShell process was spawned from PowerShell.

Sysmon recorded the process creation and Wazuh generated Rule 92027.

4. Timeline

Stage	Evidence	Result
Authentication	Windows Event ID 4624, Logon Type 10	Successful RDP observed
Post-RDP Execution	Sysmon Event ID 1	PowerShell process creation
Process Relationship	ParentImage = PowerShell; Image = PowerShell	PowerShell spawned PowerShell
Detection	Wazuh Rule 92027, Level 4	Alert generated

5. Detection Evidence

Wazuh: Rule 92027; level 4; PowerShell process-spawn detection; agent Win-11-SOC.

Field	Observed Value
UtcTime	2026-08-23 17:04:05.375
Event	Process Create
Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
CommandLine	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
User	WIN11\ vboxuser
ProcessId	6376
ParentProcessId	8392
ParentImage	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
ParentCommandLine	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
ParentUser	WIN11\ vboxuser
TerminalSessionId	1
IntegrityLevel	High

6. Technical Analysis

The telemetry shows PowerShell running as WIN11\vbouser with High integrity. Its parent process is also PowerShell under the same user context. This confirms a PowerShell-to-PowerShell process relationship after the RDP authentication stage.

The behavior is useful for post-authentication threat hunting, but the supplied evidence does not show a malicious payload, persistence mechanism, or harmful command. Therefore, the appropriate assessment is suspicious activity rather than confirmed malicious execution.

7. MITRE ATT&CK Mapping

Technique	ID	Relevance
Remote Services: Remote Desktop Protocol	T1021.001	Successful RDP authentication preceded the activity.
Command and Scripting Interpreter: PowerShell	T1059.001	PowerShell was executed and spawned another PowerShell process.

8. Investigation Findings

Successful RDP activity preceded the PowerShell execution.

Sysmon successfully recorded Event ID 1.

Wazuh detected the PowerShell-to-PowerShell relationship with Rule 92027.

Both processes ran under WIN11\vbouser.

Available evidence does not establish malicious intent.

9. Response / Containment

Validate whether the RDP login was authorized and review its corresponding 4624 telemetry.

Review the PowerShell command line and surrounding process tree for additional suspicious activity.

Correlate the timestamp with Windows Security, Sysmon, and PowerShell operational logs.

If unauthorized activity is confirmed, terminate the suspicious session and isolate the endpoint according to incident-response procedures.

Preserve relevant event and process-tree evidence before remediation where feasible.

12. Evidences

	Aug 23, 2026 @ 17:04:05.3...	Win-11-SOC	Executable file dropped in folder community...	10	-	-	-	-	92213
	Aug 23, 2026 @ 17:04:05.3...	Win-11-SOC	Powershell process spawned powershell in...	4	-	-	-	-	92027

11. Final Assessment

The Home SOC Lab demonstrated a post-RDP execution scenario in which a PowerShell process spawned another PowerShell process. Sysmon captured the activity and Wazuh generated Rule 92027 at level 4. This provides sufficient evidence to document suspicious post-RDP execution and demonstrate endpoint detection and investigation capability. Based solely on the supplied evidence, the event should not be labeled confirmed malicious.